

Frame-to-Event Conversion Framework

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Background

Motivation

- ▶ Spiking Neural Networks (SNNs) work best with **asynchronous events** instead of RGB frames.
- ▶ Large RGB datasets are widely available, while **event-camera datasets are limited**.
- ▶ Goal: reuse existing RGB videos for event-based learning.
- ▶ Enable future models compatible with **both RGB and event cameras**.

Project Objective

Develop a lightweight, classical framework that converts RGB video into an event stream

$$(x, y, t, p)$$

- ▶ No deep learning
- ▶ Low computational cost
- ▶ Validated using DSEC ground truth

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Literature Review

ESIM

Simulates DVS events using **3D rendering** and **adaptive sampling** based on optical flow.

High physical realism at the expense of **high computational cost**.

v2e

Converts video into events using **Super-SloMo interpolation**.

Includes **sensor non-idealities** for more realistic event generation.

Our Approach

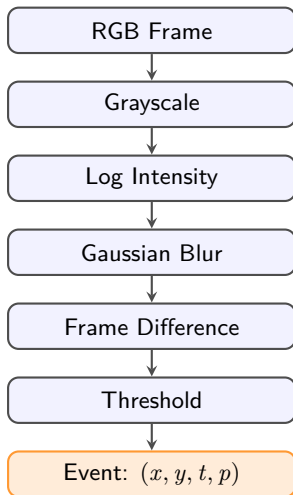
Inspired by v2e, but designed as a **lightweight classical framework**.

Uses temporal intensity differences without interpolation or sensor simulation.

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Framework Implementation



Event Generation

For every pixel, calculate the intensity change:

$$\Delta L = L_t - L_{t-1}$$

Generate an event if the change exceeds θ :

$$p = \begin{cases} +1, & \text{if } \Delta L > \theta \\ -1, & \text{if } \Delta L < -\theta \end{cases}$$

```
$ uv run python -m framework.cli \
  -i input.mp4 -o results \
  -t 0.1 --resize 640 480 --video
```

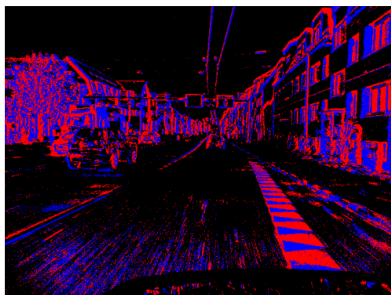
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Visual Comparison

Evaluation on **Zurich City 13** a sequence (50 ms temporal windows).

Original DSEC Event Camera



Generated Synthetic Events

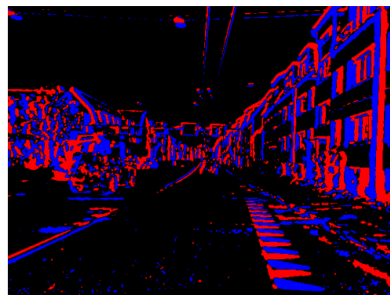
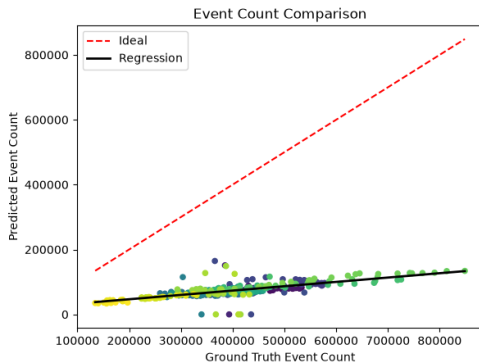


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Quantitative Results

Event Count Correlation



Pearson Correlation: **0.709**
Tracks scene activity accurately.

Sparsity Yield

18.3%

Events retained with respect to ground-truth events.

Spatial Precision

93.5%

Spatial F1-score **0.86**
Events happen in the correct regions.

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Object Detection

Tested with pre-trained **TWL SpikeYOLO**

DSEC Events (mAP: 0.18)



Synthetic Events (mAP: 0.00)



Insight: While structurally similar, the statistical distribution differs significantly. Networks trained purely on real sensors cannot generalize to synthetic data without domain adaptation or fine-tuning.

Conclusions

Achievements

- ✓ Lightweight: Avoids expensive rendering & sub-frame interpolation
- ✓ Preserves scene dynamics and spatial event locations.
- ✓ Strong correlation with real event activity ($r = 0.709$).
- ✓ Reduces redundant events while maintaining moving structures.

Limitations

- Event statistics differ from real DVS cameras.
- Zero-shot object detection does not generalize.
- Domain adaptation or fine-tuning is required for SNNs.

Thank you!

github.com/jzarcoo/Frame-to-Event-Conversion-Framework

Bibliography

-  Delbruck, T., Hu, Y., & Liu, S. (2021). *v2e: From Video Frames to Realistic DVS Events*. Institute of Neuroinformatics, University of Zürich and ETH Zürich, Switzerland.
-  Gehrig, M., Aarents, W., Gehrig, D., Scaramuzza, D. (2021). DSEC: A stereo event camera dataset for driving scenarios. IEEE Robotics and Automation Letters.
<https://doi.org/10.1109/LRA.2021.3068942>
-  Gehrig, D., Rebecq, H., & Scaramuzza, D. (2018). *ESIM: an Open Event Camera Simulator*. Robotics and Perception Group, Depts. Informatics and Neuroinformatics, University of Zurich and ETH Zurich. https://rpg.ifi.uzh.ch/docs/CORL18_Rebecq.pdf

Bibliography

-  Gehrig, D., Scaramuzza, D. (2024). Low latency automotive vision with event cameras. Nature.
-  Khitushkin, K.S., Isakov, T.T., Bakhshiev, A.V. (2026). Using Spiking Neural Networks for Event and Multimodal Data Processing in Object Detection Tasks. In: Kryzhanovsky, B., Dunin-Barkowski, W., Redko, V., Tiumentsev, Y., Klimov, V.V. (eds) Advances in Neural Computation, Machine Learning, and Cognitive Research IX. NEUROINFORMATICS 2025. Studies in Computational Intelligence, vol 1241. Springer, Cham.
https://doi.org/10.1007/978-3-032-07690-8_10